

Dietary intake of berries and flavonoids in relation to cognitive decline

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Objective Berries are high in flavonoids, especially anthocyanidins, and improve cognition in experimental studies. We prospectively evaluated whether greater long-term intakes of berries and flavonoids are associated with slower rates of cognitive decline in older women. **Methods** Beginning in 1980, a semi-quantitative food frequency questionnaire was administered every four years to Nurses' Health Study participants. In 1995–2001, we began measuring cognitive function in 16,010 participants, aged ≥ 70 years; follow-up assessments were conducted twice, at two-year intervals. To ascertain long-term diet, we averaged dietary variables from 1980 through the initial cognitive interview. Using multivariable-adjusted, mixed linear regression, we estimated mean differences in slopes of cognitive decline by long-term berry and flavonoid intakes. **Results** Greater intakes of blueberries and strawberries were associated with slower rates of cognitive decline (e.g., for a global score averaging all six cognitive tests, for blueberries: $p\text{-trend}=0.014$ and mean difference=0.04 [95% CI=0.01, 0.07] comparing extreme categories of intake; for strawberries: $p\text{-trend}=0.022$ and mean difference=0.03 [95% CI=0.00, 0.06] comparing extreme categories of intake), after adjusting for multiple potential confounders. These effect estimates were equivalent to those we find for approximately 1.5 to 2.5 years of age in our cohort, indicating that berry intake appears to delay cognitive aging by up to 2.5 years. Additionally, in further supporting evidence, greater intakes of anthocyanidins and total flavonoids were associated with slower rates of cognitive decline ($p\text{-trends}=0.015$ and 0.053 , respectively, for the global score). **Interpretation** Higher intake of flavonoids, particularly from berries, appears to reduce rates of cognitive decline in older adults.